English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, CS Year 3, 2020/2021 Test 1: solutions — *Your grade is 2 + the number of points. Time allowed: 3 hours. Good luck! Work on* $(\Omega, \mathcal{F}, \mathbb{P})$.

Problem 1. (1 point) At a shooting gallery, a boy pays 2 BGN for three shots. The prizes are 10 BGN for three hits, 5 for two hits, and 1 for one hit. His hit probability per shot is 1/3. Is the game fair?[1] For which values of the largest prize is the game favourable to him?

Solution. (0.6 points for the main part + 0.1 for $y > 12$.) Let the largest prize be y. Then

$$\begin{aligned}\mathbb{E}X = &-2\binom{3}{0}(2/3)^3 - \binom{3}{1}(1/3)(2/3)^2 \\ &+ 3\binom{3}{2}(1/3)^2(2/3) + (y-2)\binom{3}{3}(1/3)^3 = \frac{y-2}{27}.\end{aligned}$$

Thus the game is favourable for $y > 12$, unfavourable for $y < 12$, and in particular is not fair when $y = 10$. If the boy makes a profit, what is the probability he hit the target on his third shot?

Solution. (0.3 points.) Write Y for a hit and N for a miss. The probability of missing the third shot is

$$\mathbb{P}(YYN \mid \{YYN, YNY, NYY, NNN\}) = \frac{2/27}{3 \cdot 2/27 + 1/27} = \frac{2}{7}.$$

The answer is therefore $1 - 2/7 = 5/7$.

Problem 2. The first box contains 4 white and 6 black balls; the second 7 white and 3 black. Draw one ball randomly from each, add one white ball, and place all three in a third box.

1. (0.4 points) What is the probability that a random ball from the third box is white?

   Solution. Let A mean a white draw from the third box, and B_{xy} mean colours x, y from the first two, with W for white and B for black. Total probability gives

   $$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \mid B_{WW})\mathbb{P}(B_{WW}) + \mathbb{P}(A \mid B_{BW})[\mathbb{P}(B_{BW}) + \mathbb{P}(B_{WB})] + \mathbb{P}(A \mid B_{BB})\mathbb{P}(B_{BB}) \\ &= 1 \cdot \frac{4 \cdot 7}{100} + \frac{2}{3}\frac{4 \cdot 3 + 6 \cdot 7}{100} + \frac{1}{3}\frac{6 \cdot 3}{100} = \frac{7}{10}.\end{aligned}$$

2. (0.6 points) Draw from the third box with replacement until the first white ball. Let X count draws. Find $\mathbb{E}X$. Is X geometric?

   Solution. (0.3 points if all calculations are otherwise correct but use a geometric law with the probability from part 1.) With three white balls the first draw succeeds. With two white and one black, $X \sim \mathrm{Ge}(2/3)$ conditionally and its mean is 3/2. With one white and two black the conditional law is $\mathrm{Ge}(1/3)$, with mean 3. Thus

   $$\begin{aligned}\mathbb{E}X &= \mathbb{E}[X \mid B_{WW}]\mathbb{P}(B_{WW}) + \mathbb{E}[X \mid B_{BW}][\mathbb{P}(B_{BW}) + \mathbb{P}(B_{WB})] + \mathbb{E}[X \mid B_{BB}]\mathbb{P}(B_{BB}) \\ &= \frac{28}{100} + \frac{3}{2}\frac{54}{100} + 3\frac{18}{100} = \frac{163}{100}.\end{aligned}$$

   If X were geometric, part 1 would require $\mathrm{Ge}(7/10)$ and mean 10/7, contrary to this calculation.

Problem 3. (1 point) A bag has $n \in \mathbb{N}$ coins with values $1, \ldots, n$. A person takes a handful, with each coin selected with probability $p \in (0, 1)$. Find the expected sum S.

Solution. (0.1–0.2 or more for partial reasoning.) Let X_k indicate selection of coin k. Then $X_k \sim \mathrm{Ber}(p)$ and $\mathbb{E}X_k = p$. Linearity gives

$$\mathbb{E}S = \mathbb{E}\left[\sum_{k=1}^{n} kX_k\right] = \sum_{k=1}^{n} k\mathbb{E}X_k = \frac{pn(n+1)}{2}.$$

Problem 4. 1. (0.4 points) State when events $A, B \subset \Omega$ are independent and when they are disjoint. Can they be independent if (a) disjoint, or (b) $A \subset B$? If so, under what conditions?

Solution. Award 0.1 for each definition and each example. Disjoint means $A \cap B = \emptyset$; independent means $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Both answers are yes: (a) requires $\mathbb{P}(A) = 0$ or $\mathbb{P}(B) = 0$; (b) requires $\mathbb{P}(A) = 0$ or $\mathbb{P}(B) = 1$.

[1] A game is fair if expected net winnings are zero.

2. (0.6 points) State when discrete random variables X, Y are independent and when identically distributed.

   Marking. 0.1 per definition, 0.1 for $\mathbb{P}(X < Y) = 1/2$, 0.1 for inclusion–exclusion, with at most 0.3 for these first four items; 0.3 for the last part.

   Solution. Identical distributions mean $\mathbb{P}(X = a) = \mathbb{P}(Y = a)$ for every $a \in \mathbb{R}$. Independence means $\mathbb{P}(X = a, Y = b) = \mathbb{P}(X = a)\mathbb{P}(Y = b)$ for all a, b. Let X, Y, Z be identically distributed and pairwise independent. Assume ties have probability zero. Find $\mathbb{P}(X < Y)$, $\mathbb{P}(Y < Z)$ and $\mathbb{P}(Z < X)$.

   Solution. By independence and identical laws, $\mathbb{P}(X < Y) = \mathbb{P}(X > Y)$. Since ties have probability zero and the three comparison probabilities sum to one, each requested probability is $1/2$. State inclusion–exclusion for three events.

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).\end{aligned}$$

   Prove $a = \min\{\mathbb{P}(X < Y), \mathbb{P}(Y < Z), \mathbb{P}(Z < X)\} \leq 2/3$.[2]

   Solution. Set $A = \{X < Y\}$, $B = \{Y < Z\}$, $C = \{Z < X\}$. Their triple intersection is empty. Their pairwise intersections are disjoint, so

$$\mathbb{P}(A \cap B) + \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C) = \mathbb{P}((A \cap B) \cup (A \cap C) \cup (B \cap C)) \leq 1.$$

   Inclusion–exclusion gives $1 \geq 3a - 1$, as required.

   This illustrates how, whichever of three candidates wins, two thirds of voters may prefer another candidate. Three voters can have rankings $A > B > C$, $B > C > A$, $C > A > B$. In probabilistic notation, $\mathbb{P}(X = 1, Y = 2, Z = 3) = \mathbb{P}(X = 2, Y = 3, Z = 1) = \mathbb{P}(X = 3, Y = 1, Z = 2) = 1/3$. A related observation dates to Condorcet in the late eighteenth century; see the Condorcet paradox.

[2] Can you construct equality? A simple example uses suitable probabilities on events of the form $(X = n_1, Y = n_2, Z = n_3)$.